BoardBuddy testing

Test Prompts

Prompts with specific right-or-wrong answers

When was the last time MWD implemented the WSAP?
Correct answer (2015) **A+**

...and before that? (followup)
Correct answer (2009 and 2010) **A+**

What is MWD's current Tier 1 supply rate?
Correct but slightly out-of-date response ($321/AF for 2023). **B+**

What was MWD's $/sqft incentive for turf removal in 2013-18?
Only cited 1 source, from 2025. **B+**

What are Metropolitan's climate goals?
Cited 40% reduction by 2023 relative to 1990, and carbon neutrality by 2045. Provided three relevant sources, but not the Climate Action Plan. **A**

Prompts requiring synthesis across multiple sources

Tell me about the history of MWD's land acquisitions in the Palo Verde Valley
Briefly described the three major land purchases in a bulletted list, pulled from the PV White Paper. **A+**

Tell me about the history of MWD's policies and programs related to stormwater capture and recharge
Gave an adequate summary of the Stormwater Pilot Program, but lacking in detail and in a poorly-formatted list. **B+**

What are the components of MWD's rate structure?
Gave a brief description of the six components. Notes that the Water Stewardship Rate ceased after 2020. **A+**

What names have been given to the Delta tunnels project over the years?
Indicated the BDCP, WaterFix, and DCP and gave a short description of each. Did not provide dates until prompted to do so in a followup. **A**

Give a detailed history of MWD's rate structure over the decades.
Provided an adequate but terse summary of the major changes since MWD's foundation, in an excessively bullet-pointed list. Provided a more detailed narrative summary when prompted with a followup. But only two sources were cited, both from 2023. **B**

What events precipitated the signing of the QSA?
Gave an extremely brief response that cited increases in AZ and NV's use during the 1990s and "a severe drought", with a single 2024 source. **C**

Please return any Board letters that approved or modified HECW rebates or water savings estimates.
(suggested by teammate Lindsay who was recently frustrated by the inability to easily search the Board letter archive)
Did not understand what was meant by 'HECW'. After clarifying, it found no board letters referencing clothes washers, but returned a few letters related to other conservation programs. **F**

General comments

The sourcing is excellent. Nearly all sources cited are relevant to the request. And they are linked to the original documents!

It does an excellent job finding specific information when that information can be found in one or two sources.

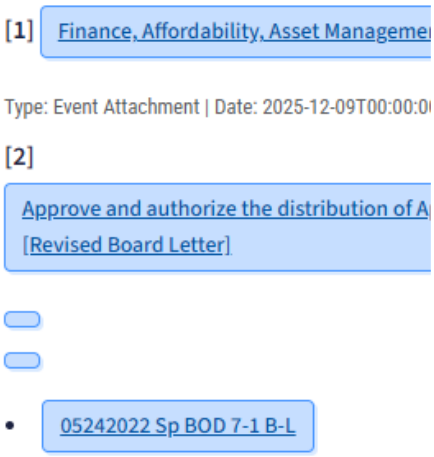
It does a pretty good job of summarizing information across multiple sources, however it seems to be limited by the small number of sources that it cites. Not sure if this is due to the current archive of material being historically limited, or parameters of the semantic search limiting the number of chunks returned. This results in responses that are often terse, lacking in detail and context. With more sources, I get the sense that it would be able to create richer responses.

I did not encounter any noticeable hallucinations or obvious wrong answers.

After using this prototype I am convinced that a more functional version would be incredibly useful to staff.

Bugs

1. **Dark mode contrast**
2. **Redundant source links**. Often, sources show additional blue bubbles which all link to the same document. e.g. all four bubbles in source #2 below link to the same document:



Suggestions

1. **More verbose responses.** Currently it gives very concise, matter-of-fact responses. This may be a symptom of too little source material to draw from. I rarely feel this way about LLM responses, but I would actually prefer a bit more detail, *if it exists in the source documents*, so long as that doesn't introduce more hallucinations. Perhaps the system prompt can also be modified to encourage this.
2. **More sources**. In testing, the model usually cited one to three sources, seeming to prefer more recent sources. Even given a very general prompt ('Describe the history of the Pure Water project') it never cited more than five sources.
3. **Reduce excessive use of hierarchical bullet points**. The model seems to respond with bullet point lists by default, often with excessive nesting of up to 5 levels. I would prefer a narrative form with headings in many cases, though this is user preference. Even when asked for a narrative, it tends to return lists. Example of excessive nesting:

• Launch of the Stormwater Pilot Program

- In 2019, MWD approved a dedicated stormwater pilot program with a total funding commitment of $12.5 million.
  - The program was structured with two components:
    • A Direct Use pilot component ($5 million)
    • A Recharge pilot component ($7.5 million) - The purpose was to evaluate the water supply benefits of capturing stormwater—both for direct distribution and for enhancing groundwater recharge. - An application period ran from January 2020 to December 2021. Although 12 applications were submitted (with 4 later withdrawing), 6 projects ultimately received approval [2].

• Program Goals and Achievements

4. **A 'New Conversations' button** would be helpful to avoid contaminating the context window with previous requests.
5. **Ensure that public documents are referenced**. I did not see it cite any public documents, even when asked about them ('Tell me about the Climate Action Plan')
6. **Option to display semantically-relevant chunks from source documents**. Put this in the 'nice-to-have but not necessary' category.